Certified Shortcuts and Depth Bounds for Full Optimal Binary Search Trees

Research checkpoint: exact shortcuts and barriers to relaxations

OpenAI Codex Sydney

Problem selection and sponsorship: the repository owner

October 18, 2026

Abstract

We investigate exact construction of optimal binary search trees with weights on both keys and gaps. We give sufficient conditions for contracting a single key, or a contiguous block of keys and gaps, without changing the optimum except for an explicitly known additive cost. A linear preprocessing pass selects certified blocks from a Cartesian-tree hierarchy and chooses a size threshold minimizing the number of subsequent dynamic-programming states. If the surviving core has m keys and the selected blocks have sizes k_t, construction takes $O(n + m^2 + \sum_t k_t^2)$ arithmetic operations. The method remains quadratic in the worst case. On one seeded positive-weight instance with 512 keys and sparse heavy separators, it reduced the number of DP states from 131,328 to 10,344; ordinary random mixed weights benefited little. An additional certificate optimizes all minimum-unweighted-path-length shapes and excludes every other shape using a residual-weight lower bound. A dyadic extension also optimizes trees with bounded excess unweighted path length and certifies them against all remaining shapes. The shortcuts provide an $O(n \log n)$ exact route when their explicit tests pass, including tested near-uniform instances left unresolved by contraction. We also document an order-free Huffman relaxation, its failed sparse-DP implementation, and exact counterexamples to proposed local certificates. A second relaxation prices depth-count constraints along the ordered key sequence. An affine-tail lemma evaluates it in $O(n \log n)$ time without assuming the optimal tree is shallow. We characterize the missing ordering condition and give a strictly positive family showing that unique-minimum checks on any fixed window width cannot make this relaxation exact. These are research results and reproducible observations, not a resolution of general subquadratic construction. Novelty of the contraction conditions has not been established.

1 Model, scope, and prior work

Keys have fixed inorder positions $0, \dots, n-1$ and nonnegative weights $p_0, \dots, p_{n-1}$. The $n+1$ gaps have weights $q_0, \dots, q_n$. A key at depth d contributes $p(d+1)$; a gap at depth d contributes qd. Charging an extra comparison at each gap adds the tree-independent constant $\sum q_i$.

For a half-open key interval $[i, j)$, write

$$W(i,j) = \sum_{r=i}^{j-1} p_r + \sum_{r=i}^{j} q_r, \qquad C(i,i) = 0.$$

The exact recurrence is

$$C(i,j) = W(i,j) + \min_{i \le r < j} \{C(i,r) + C(r+1,j)\}.$$

Knuth's monotone-root optimization computes the table in $O(n^2)$ time [1]. The central obstacle here is the number of states, not merely the cost per state. We retain both kinds of weights throughout.

Larmore gives subquadratic exact construction under restrictions on long runs of very low frequency items, together with an approximation tradeoff [2]. Karpinski, Larmore, and Rytter combine exact small intervals with a bounded top-level construction [3]. These results are essential context: an improvement on a restricted family must not be described as the first subquadratic result. This checkpoint does not make a comprehensive priority claim. The conditions below were derived in this investigation; a targeted literature search did not establish whether equivalent rules already exist.

All running-time bounds below count arithmetic operations and comparisons. They are not bit-complexity bounds for unbounded integer weights.

2 A sharper single-key contraction

A key is a *cherry* when its two children are external nodes. Replacing a cherry with one gap is exact: if the key and its gaps have weights p, x, y, their total is $s = p + x + y$. A cherry rooted at depth d contributes $s(d+1)$, whereas the replacement gap contributes sd.

Theorem 1 (Single-key rule)**.** *Let a key of weight p have adjacent gap weights x, y. If its immediate predecessor and successor keys have weights a, b and*

$$\min(a, b) \;\geq\; 2p + x + y,$$

there is an optimal tree in which this key is a cherry. At the left endpoint, it suffices that the successor weight is at least $p + x$; at the right endpoint, it suffices that the predecessor weight is at least $p + y$. Consequently contraction to a gap of weight $s = p + x + y$ satisfies $\mathrm{OPT}(\textit{original}) = \mathrm{OPT}(\textit{contracted}) + s$.

Proof. Consider the key at depth d in any tree. If its left subtree is nonempty, the immediate predecessor has depth $d + l$, $l \geq 1$. If its right subtree is nonempty, the immediate successor has depth $d + r$, $r \geq 1$. Set the corresponding distance to zero for an empty internal side.

When $l \geq 1$, move the predecessor to the chosen key's position, replacing the predecessor's old position by its left child. Insert the chosen key as a cherry at the leftmost gap of the former right subtree. Inorder order is preserved. All other outcomes either retain their depths or become shallower. The predecessor saves la, and the cost change is at most

$$-la + (r+1)p + (r-l+1)x + y. \tag{1}$$

The mirrored successor exchange has change at most

$$-rb + (l+1)p + x + (l-r+1)y. \tag{2}$$

If both sides are nonempty and $l \geq r$, substituting $a \geq 2p + x + y$ in (1) gives

$$(r + 1 - 2l)(p + x) + (1 - l)y \leq 0.$$

Use (2) when $r \geq l$. If only the left side is nonempty, $r = 0$ and $a \geq p + y$ makes (1) nonpositive. The other case is symmetric. The stated interior condition implies these one-sided conditions. At a global endpoint only the appropriate one-sided case is possible. With both sides empty the key is already a cherry.

Applying the exchange to an optimum proves the existence claim. Contraction subtracts exactly s. Conversely, every contracted tree expands to a valid original tree at additional cost s, proving equality. □

The weaker condition $\min(a, b) \geq p + x + y$ is false. For $p = (24, 14, 24)$ and $q = (1, 0, 9, 0)$, the middle key has local mass 23 and neighbors of weight 24. Contracting it forces cost 142, while the unrestricted optimum costs 130.

Maintain a doubly linked list of remaining keys and weighted gaps. Initially queue every key; after a contraction, only its predecessor and successor need rechecking. At most $3n$ queue entries are processed. Saving the contracted tree at the new gap permits reconstruction. Thus this pass takes $O(n)$ operations and space, and a Knuth solve on the remaining m keys gives total time $O(n + m^2)$. No confluence or minimum-core-size property is claimed.

3 Whole-block contraction

Let $B = [i, j)$ be a nonempty proper interval, including its endpoint gaps q_i, q_j. Put $W = W(i, j)$ and $C = C(i, j)$, with costs measured from a stand-alone root of depth zero.

Theorem 2 (Block rule)**.** *If both boundary keys exist and their weights satisfy*

$$p_{i-1} \geq W + C, \qquad p_j \geq W + C,$$

there is an optimum in which all of B forms one subtree. For a prefix or suffix block, it suffices that the sole boundary key has weight at least C. Contracting the block to one gap of weight W then gives

$$\mathrm{OPT}(\textit{original}) = \mathrm{OPT}(\textit{contracted}) + C.$$

Proof. In any tree, let v be the lowest common ancestor of gaps q_i, q_j, at depth d. Its key lies in B, and every outcome of B descends from v. Hence the original contribution of B is at least $(d+1)W$.

Within the subtree rooted at v, delete the keys of B from the inward fringes of its two children. On the left, a deleted key has no retained right child and can be suppressed in favor of its left child; on the right the process is symmetric. Let the retained outside trees be A, D. All their retained outcomes are at depths no greater than before. If A is nonempty, its maximum is boundary key $i - 1$; if D is nonempty, its minimum is boundary key j.

Suppose both are nonempty, and their boundary keys had original depths $d + l, d + r$. If $l \geq r$, remove the maximum key from A, replacing its position by its left child, and make that key the new root at v. Its left child is the remainder of A and its right child is D. Put an optimal tree for B at the leftmost external position of D. This position has depth at most $d + r + 1$. No outside outcome other than the promoted boundary moves deeper, and that boundary saves lp_{i-1}. The increase in the contribution of B is at most

$$C + (d + r + 1)W - (d + 1)W = C + rW.$$

Since $l \geq r \geq 1$ and $p_{i-1} \geq W + C$, the saving is at least $l(W + C) \geq rW + C$. The mirrored construction applies for $r \geq l$.

If only one retained outside tree is nonempty, promote its inward boundary key to v and attach the optimal block as the other child. The block increase is at most C, while that boundary saves at least its own weight. The interior hypothesis suffices, and for a global prefix or suffix the weaker stated hypothesis suffices. If both outside trees are empty, B already occupies the subtree at v and can be optimized in place.

The construction never increases cost. Apply it to an optimum. A block at depth e contributes $C + eW$, while its replacement gap contributes eW. Contraction and expansion now prove the equality. □

A balanced k-key tree has height $h_k = \lceil \log_2(k+1) \rceil$ and cost at most $h_k W$. Thus the interior test

$$\min(p_{i-1}, p_j) \geq (1 + h_k)W$$

and endpoint test $p_{\text{boundary}} \geq h_k W$ certify a block without first solving it. These are sufficient conditions, not characterizations.

4 An adaptive linear selection pass

Build the maximum Cartesian tree of the key-weight sequence, breaking ties deterministically. Its subtree intervals form a laminar family: intervals are disjoint or nested. Compute their sizes and masses by one traversal and prefix sums. Test each proper interval with the balanced-tree bound above. This inspects only $O(n)$ candidates. The restriction to Cartesian intervals is an algorithmic choice, not a claim that all useful blocks occur there.

For a threshold b, select the maximal certified intervals whose sizes are at most b. Different selected intervals have a surviving separator key between them. Contracting one therefore leaves every other selected interval's weights and boundary-key weights unchanged, so the block theorem applies successively. Write their sizes as $k_1, \ldots, k_s$, and $m = n - \sum_t k_t$. The exact number of nonempty interval-DP states is

$$S(b) = \tau(m) + \sum_t \tau(k_t), \qquad \tau(u) = \frac{u(u+1)}{2}.$$

We choose a threshold minimizing $S(b)$ without running these DPs. For every certified interval I of size k, find its maximal proper certified descendants J. Immediately before I becomes eligible, these descendants are its selected blocks. Its activation therefore changes the core size and block-state total by

$$\Delta m = -\left(k - \sum_J |J|\right), \qquad \Delta S_{\text{blocks}} = \tau(k) - \sum_J \tau(|J|).$$

These sums are computed bottom-up. Bucket activation events by integer size $1, \ldots, n$ and scan the buckets, maintaining m and the block-state total. There are $O(n)$ events and buckets. Equal-size intervals are disjoint; evaluate a threshold after processing its entire bucket. The empty selection $b = 0$ is included.

Corollary 3. *The selected threshold and blocks are found in $O(n)$ arithmetic operations and space. Solving the blocks and surviving core by Knuth's method and expanding their saved trees takes*

$$O\left(n + m^2 + \sum_t k_t^2\right)$$

time and $O(n + m^2 + \max_t k_t^2)$ space when block tables are released after each solve. The chosen state count is no larger than $n(n+1)/2$. Worst-case time remains $O(n^2)$.

The minimization is only over these size thresholds in this candidate hierarchy, not over every possible partition. It does not minimize wall-clock time. Preprocessing, table allocation, root scans, and reconstruction still count as work.

For perspective, a certified partition with $m = O(n^{2/3})$ surviving keys and every $k_t = O(n^{1/3})$ gives $O(n^{4/3})$ time, since $\sum k_t^2 \le (\max k_t)n$. This is a conditional structural consequence, not an unrestricted bound.

5 Recursive construction with a bounded preprocessing depth

The implementation `recursive_contract.py` combines the proved components. At each call it applies the single-key pass, compares the order-free lower bound of Section 9 with a feasible bisection-tree cost, and returns immediately on equality. Otherwise it selects blocks as above. It recurses only if their combined table-state estimate is at most three quarters of the full estimate; small components and unsuccessful decompositions fall back to Knuth's method.

Suppose a component has $m > 8$ keys after the single-key pass and is successfully decomposed. Every child component of size s satisfies $s(s+1) \le \frac{3}{4}m(m+1)$, hence $s < 0.92m$. Child sizes sum to m. Consequently the recursion has logarithmic depth, and the total number of keys processed on any recursion level is at most n. Linear selection and reconstruction cost $O(n \log n)$ in total. The heap-based equality checks cost $O(n \log^2 n)$ over all levels. If the terminal DP sizes are r_t, the construction therefore takes

$$O\left(n \log^2 n + \sum_t r_t^2\right)$$

arithmetic operations. Its worst case is still quadratic. This threshold prevents repeated expensive scans after arbitrarily small reductions. Correctness follows by contraction, exact equality certificates, and induction on the strictly smaller recursive components; reconstruction restores the original key indices and gap subtrees.

The recursive implementation passed 3,000 further random inputs through $n = 30$ against the cubic oracle and direct tree evaluation. A separate nested fixture checks composition across multiple decomposition levels. Matched Python timings use five inputs per family, alternate which solver runs first, and include preprocessing and reconstruction for both solvers. At $n = 1024$, the median runtime ratio (recursive divided by Knuth) was 0.134 on the positive-weight heavy-separator family; its median terminal DP count was 30,017 versus 524,800. On random mixed weights the ratio was 0.986 and on near-uniform weights 1.001, which support no meaningful gain. At $n = 512$, the respective random and near-uniform ratios were 1.131 and 1.093: preprocessing can make the method slower. These are exploratory measurements on one M4 host, not production-language or cross-machine claims.

6 A certified shortcut through nearly complete shapes

Let $I(T)$ be the sum of internal-node depths, with root depth zero. The external path length is $I(T) + 2n$, by induction on the two subtrees. Consequently uniform key weight a and gap weight b give cost

$$(a + b)I(T) + (a + 2b)n.$$

For an arbitrary instance set $a = \min_i p_i$, $b = \min_j q_j$, and let H' be the order-free mixed-node Huffman bound of Section 9 on residual weights $p_i - a, q_j - b$. These weights remain nonnegative. Write $I_*(n)$ for the minimum possible unweighted internal path length, and let C_* be the best weighted cost among all trees with $I(T) = I_*(n)$.

Theorem 4 (Minimum-path-length certificate)**.** *For $n \geq 1$, if*

$$C_* \;\leq\; (a + b)(I_*(n) + 1) + (a + 2b)n + H',$$

then a tree attaining C_* is a globally optimal full OBST.

Proof. Split each outcome weight into its uniform baseline and residual. Every tree outside the minimum-I class has integer $I(T) \geq I_*(n) + 1$, and its residual cost is at least H'. Its full cost is therefore at least the right-hand side. Within the class, C_* is minimal by definition. Equality is sufficient, including when $a + b = 0$. □

Optimize the class without interval tables. Put $h = \lfloor \log_2(n + 1) \rfloor$, $S = 2^h$, $m = S - 1$, and $r = n - m$. Every minimum-I tree has the complete upper skeleton with m keys, and expands exactly r of its S external slots into single-key cherries. Indeed, a missing shallower slot could otherwise receive a deepest internal leaf, strictly reducing I. Inorder relabeling preserves search-tree feasibility in this unweighted shape argument. Thus

$$I_*(n) = (h - 2)2^h + 2 + hr.$$

Process skeleton slots $j = 0, \ldots, S - 1$ in inorder. State (j, e) records the number of processed slots and expansions, so the next input index is $i = j + e$. An unexpanded slot charges hq_i; an expanded slot charges $(h+1)(p_i+q_i+q_{i+1})$. Except after the last slot, also charge the next skeleton key at comparison length

$$d_j = h - v_2(j + 1),$$

using p_i after an unexpanded slot and p_{i+1} after an expansion. Here v_2 counts trailing zero bits. Exactly r expansions are required, so state (j, e) is retained only when

$$\max(0, r - (S - j)) \leq e \leq \min(r, j).$$

With $s = \min(r, S - r)$, each stage has at most $s + 1$ states and each state has two outgoing choices. This computes C_* and backpointers in $O(n(1 + s))$ time and space. A linear Cartesian-tree pass converts the chosen comparison lengths into the labeled tree. No key or gap weights are discarded in these transitions.

There is an exact early rejection test. Let F be the cost obtained by assigning independently sorted key and gap weights to the shallowest depths of a minimum-I shape. This is feasible for the order-free residual relaxation, so $H' \leq F - [(a + b)I_* + (a + 2b)n]$. Therefore $C_* > F + a + b$ proves the certificate cannot succeed. Otherwise the sorted lists are reused in a two-queue Huffman implementation: nonnegative weights and nondecreasing internal weights make the sequence of

merged weights nondecreasing. This adds linear work after sorting. The queue optimization is standard and is not claimed as novel.

The implementation attempts this DP only when $s \le \lfloor \log_2 n \rfloor + 1$. Including the residual bound, preprocessing, and reconstruction, a successful attempt costs $O(n \log n)$ arithmetic operations and space. A skipped or failed attempt falls back to Knuth, or to the preceding recursive construction. Both versions remain exact and have quadratic worst-case time. The test is a certificate, not an assumption that near-uniform inputs have balanced optima. Novelty of this shortcut has not been established.

Verification and limits. Minimum-I optimization matched independent enumeration of all tree shapes through eight keys on 360 weighted inputs. Both fallback versions matched independent cubic DP and direct tree costs on 3,000 further mixed-weight inputs through thirty keys; the unrestricted certificate succeeded on 768. At $n = 1,024$, one seeded input with every weight in $\{999, 1000, 1001\}$ used 2,048 shape states, 3,070 transitions, 10,511 sorting comparisons, and 1,350 merge comparisons, certifying cost 19,479,590. The nonminimum-I lower bound was 19,480,123. Knuth used 524,800 states and 1,050,139 root candidates. These work units have different costs; the state ratio is not a timing claim.

The selected perturbation sweep certified amplitude 8 around 1,000 at $n = 256$, amplitude 4 at $n = 512$, and amplitude 1 at $n = 1,024$; larger tested amplitudes at those sizes failed. These are seeded observations, not universal thresholds. Ordinary random inputs failed the certificate. Sizes 384, 768, and 1,000 deliberately skip the shortcut because their minimum-I shape space exceeds the logarithmic width budget.

Six matched trials per fixture used all six orderings of Knuth, recursive contraction, and the combined solver, with identical inputs and correctness checks outside timing. On Apple M4 with Python 3.9.6, median per-trial combined/Knuth time ratios at $n = 1,024$ were 0.03934 for near-uniform, 0.10478 for heavy separators, 1.05454 for random, 0.91929 for log-uniform, and 1.03117 for gap spikes. All six near-uniform cases certified. The skipped near-uniform controls at $n = 768$ and $n = 1,000$ gave ratios 1.17886 and 1.04896. At $n = 512$ the combined method was about 21% slower than recursive contraction alone on heavy separators. These exploratory timings include sorting, lower bounds, and reconstruction; they do not establish broad superiority. The final early-rejection and two-queue implementations were also checked against the original heap certificate on all 3,000 inputs.

Optimize the baseline rather than fix it. The largest subtractable baseline need not give the strongest certificate. Set $a_0 = \min p_i$, $b_0 = \min q_j$, and use $a = a_0 - \delta$, $b = b_0 - \delta$ for an integer $0 \le \delta \le \lfloor (a_0 + b_0)/2 \rfloor$. The residual weights remain nonnegative and $a + b \ge 0$. Although one baseline term may become negative, it is only an algebraic decomposition; the input probabilities are unchanged. Let H_δ denote the residual order-free optimum. The exclusion bound is

$$B(\delta) = (a_0 + b_0 - 2\delta)(I_* + 1) + (a_0 + 2b_0 - 3\delta)n + H_\delta.$$

Every tree outside the minimum-I class costs at least $B(\delta)$. Thus any sampled price with $C_* \le B(\delta)$ certifies the same candidate.

For a fixed residual tree, increasing δ adds one unit to every key and gap weight, changing its cost at rate $2I + 3n$. Consequently B is a pointwise minimum of affine functions, hence concave. A minimizing residual tree supplies a supergradient

$$g = 2(I - I_* - 1), \qquad B(x) \le B(\delta) + g(x - \delta).$$

If $g > 0$, smaller prices cannot improve the current value; if $g < 0$, larger prices cannot improve it. If $g = 0$, the value is globally maximal. Binary interval elimination, retaining the best sample, therefore finds the maximum over allowed integer prices when run to completion.

The same linear oracle supplies I. Give every original outcome secondary weight1, order queue items lexicographically by residual mass and number of original outcomes, and sum both merge coordinates. The first gives H_δ and the second gives $2I+3n$. This is the Huffman exchange argument with an infinitesimal positive common perturbation, choosing minimum I among residual-cost ties. The weight order is independent of δ, so no additional sorting is needed. Each price costs $O(n)$.

The fixed-profile ceiling extends to all prices: $B(\delta) \le F + a_0 + b_0 - 2\delta$. Therefore an input rejected by the existing profile test cannot be rescued by this search. The implementation tries prices only after the old certificate fails while passing that test, and permits at most $\lfloor \log_2 n \rfloor + 1$ extra oracle calls. This preserves the $O(n \log n)$ attempted-shortcut bound. Reaching the cap does not establish that the best price was found; every sampled certificate is still valid, and failure invokes the exact fallback.

The lexicographic oracle matched 400 exhaustive shape-profile comparisons through eight keys. Uncapped search matched exhaustive evaluation of all allowed integer prices on 1,200 inputs. On 3,000 additional full-OBST inputs through thirty keys, both capped fallback versions matched cubic DP and direct tree costs; 87 previously unsuccessful unrestricted certificates were recovered. A separate seeded sweep recovered the $n = 1,024$, amplitude 2 input with two additional oracle calls, and the $n = 2,048$, amplitude 1 input with three. Larger tested perturbations usually exceeded the profile ceiling. This applies standard concave-price optimization to the certificate; it is not a general subquadratic result.

In the seeded 2,048-key case, the candidate costs 43,032,446; the bound rises from 43,031,347 at $\delta = 0$ to 43,032,459 at $\delta = 125$. The three extra passes use 6,144 contractions and 6,141 pair comparisons, in addition to 4,096 shape states, 6,142 transitions, 20,784 sorting comparisons, and 2,752 original merge comparisons. Knuth uses 2,098,176 interval states and 4,198,710 root candidates on that input.

A separate six-trial comparison uses all orderings of Knuth, the old hybrid, and the price-search hybrid. All six $n = 1,024$, amplitude 2 cases and all six $n = 2,048$, amplitude 1 cases were newly certified. Median per-trial price-search/Knuth ratios were respectively 0.04953 and 0.06128. At $n = 2,048$, amplitude 2 still failed, with ratio 0.99391. Ordinary random and gap-spike controls at $n = 1,024$ gave ratios 1.06492 and 1.12941. The already-certified $n = 1,024$, amplitude 1 control made no extra oracle calls yet showed a price-search/old-hybrid ratio of 1.13564. This variability limits interpretation of small timing differences; operation counts and all raw observations are retained. The host was Apple M4 with Python 3.9.6.

7 Exact bounded-excess construction by dyadic intervals

The candidate class can be expanded beyond minimum internal path length. This is necessary even on some mildly nonuniform inputs. For $n = 255$, set all gap weights to 16 and all key weights to 16, except keys $8j + 1$ which have weight 17. The unique minimum-I tree costs 61,680; Knuth gives optimum 61,656, attained with $I = I_* + 2$. The algorithm below constructs and certifies this optimum. It retains both kinds of weight and never assumes that the restricted optimum is globally optimal.

An exact representation. An ordered full binary tree is a tiling of $[0, 1]$ by consecutive aligned dyadic intervals: a gap at depth d has width 2^{-d}. A leaf starting at x must satisfy $x/2^{-d} \in \mathbb{Z}$.

Conversely such a tiling is a complete prefix code and determines an ordered full tree. Alignment, not just the Kraft equality, enforces order. On a grid of height H, an interior boundary at integer coordinate Y corresponds to a key with comparison length $H - \nu_2(Y)$. Thus gap and key costs are both local in a left-to-right tiling construction.

Theorem 5 (Bounded-excess construction)**.** *For $n \geq 1$, let $N = n + 1$, $h = \lfloor \log_2 N \rfloor$, $S = 2^h$, $r = N - S$, and $s = \min(r, S - r)$. For any nonnegative integer budget k, the minimum full weighted cost among all trees with $I \leq I_* + k$ can be found and reconstructed in*

$$O\big(n(s + k + 1)(k + 1)^2 2^k\big)$$

arithmetic operations, using $O\big(n(s + k + 1)(k + 1)2^k\big)$ space for backpointers.

Proof. Measure dyadic widths in units of 2^{-h} and put $w = 2^{h-d}$. For integral depth d, the potential

$$f(d) = d - h - 2 + 2^{h+1-d} = d - h - 2 + 2w$$

is nonnegative, with zeros exactly at h and $h + 1$. The minimum external path length is $hN + 2r$. Since external path length equals $I + 2n$ and $\sum_i w_i = S$, every full tree satisfies

$$\sum_i f(d_i) = \sum_i d_i - (h + 2)N + 2S = I - I_*.$$

Every depth in the budgeted class therefore has $f(d) \leq k$. This implies $d \leq H = h + k + 2$, and only $O(k + 1)$ integral depths are possible. For integral d, direct substitution gives

$$\tfrac{1}{2} - \tfrac{1}{2} f(d) \leq w \leq 1 + f(d).$$

Apply these inequalities to a prefix of i leaves and to its remaining suffix, each having potential at most k. The cumulative position y must obey

$$\max\left(\tfrac{i-k}{2}, i - r - k\right) \leq y \leq \min\left(i + k, \tfrac{S-r+i+k}{2}\right).$$

This interval has width at most both $r + 2k$ and $(S - r)/2 + k$, hence $O(s + k)$.

Set $K = 2^{k+2}$. A state after i leaves stores integral drift $u = K(y - i)$ and spent potential $z = K \sum f(d)$. For depth d, let $W = 2^{H-d}$. Require $(iK + u) \bmod W = 0$ and update

$$u' = u + W - K, \qquad z' = z + K(d - h) + 2(W - K).$$

Retain the position band and $0 \leq z' \leq Kk$. Charge $q_i d$ and, for $i > 0$, charge $p_{i-1}(H - \nu_2(iK + u))$. Start at $(u, z) = (0, 0)$ and finish after N leaves with $u = -rK$. Every retained complete path is an aligned full tiling with excess at most k; every tree in the class passes the same tests. The DP therefore optimizes exactly this class. For fixed u, the congruence $z \equiv 2u \pmod{K}$ leaves at most $k + 1$ possible spent potentials. There are $O((s + k + 1)2^k)$ positions per stage and $O(k + 1)$ choices per state, proving the bounds. The bands admit direct array indexing; the Python prototype uses sparse dictionaries. Backpointers recover gap depths; the boundary lengths reconstruct the tree in linear time. □

Perfect-size refinement. When $N = S$, use instead $f_0(d) = d - h + \frac{3}{2}(2^{h-d} - 1)$. It is nonnegative on integer depths, vanishes only at h, and also sums to $I - I_*$. The inequality $|2^{h-d} - 1| \leq 2 f_0(d)$ and zero total drift imply $|y - i| \leq k$. Here $H = h + k + 1$, $K = 2^{k+1}$, and $z = 2K \sum f_0$. Its increment is $2K(d - h) + 3(W - K)$, and $z \equiv 3u \pmod{2K}$. The implementation uses this smaller grid at perfect sizes, with the same asymptotic bound at $s = 0$.

Certify against all other trees. Let C_k be the computed candidate, $a = \min p_i$, $b = \min q_i$, and H' the order-free bound on residual weights. Every excluded tree has $I \geq I_* + k + 1$, so it costs at least

$$B_k = (a + b)(I_* + k + 1) + (a + 2b)n + H'.$$

Thus $C_k \leq B_k$ proves global optimality. Sorting and the two-queue bound cost $O(n \log n)$ and are included in the counters. The constructor uses $k = 2$ and attempts the shortcut only when $s \leq \text{bit_length}(n)$. Its attempted work is $O(n \log n)$; a failed certificate or skipped size falls back to Knuth. The total worst case remains quadratic. Arbitrary growth of k is exponential, and wide shape spaces remain quadratic even for fixed k.

Validation, failed precursor, and prior work. The implementation passed 1,200 restricted-optimum comparisons against all shapes through eight keys with budgets $0, \ldots, 4$, and 3,000 full mixed-weight checks against cubic DP. It separately checks direct tree costs, both residual-bound implementations, and exact fallback. An additional 104 family checks include zeros, geometric weights, spikes, and key-only/gap-only controls. The seeded size sweep is retained. A precursor substituting ordered depth bounds into the old certificate passed 480 exhaustive checks but rescued none of 24 near-uniform fixtures at its 32-iteration budget. This negative experiment did not prove optimal multiplier search useless; its cost and the failed perfect-tree hypothesis motivated expanding the candidate class.

On the 1,023-key periodic fixture with weights 16 and 17, the excess-two DP finds cost 312,154 and excludes other shapes with bound 312,185. It uses 27,570 states, 137,835 transition checks, 6,052 sorting comparisons, and 1,022 merge comparisons, versus Knuth's 523,776 states and 1,046,532 root candidates. At $n = 1{,}024$, a seeded amplitude-four perturbation uses 48,003 states and 240,000 transition checks, plus 13,784 sorting and 1,264 merge comparisons. It certifies cost 19,477,996.

Six matched trials per fixture use all orderings of Knuth, the preceding baseline-price hybrid, and the excess-two constructor. Median excess-two runtime divided by Knuth runtime was 0.23600 and 0.44831 on amplitude-four perturbations at $n = 1{,}023$ and 1,024, and 0.20790 and 0.17101 on the periodic fixture at $n = 1{,}023$ and 2,047. Each certified in all six trials. Ordinary random input at $n = 1{,}023$ failed and cost 1.34949 times Knuth; amplitude-sixteen input at $n = 2{,}047$ failed and cost 1.13631 times Knuth. The skipped-width control at $n = 768$ ran the same Knuth routine and measured 0.93148, illustrating noise rather than a computational gain. These are exploratory M4/Python timings including all preprocessing and reconstruction. Periodic fixtures are repeated identically; random and perturbation inputs vary between trials. All 42 observations are retained.

The dyadic/prefix-code viewpoint and sequential frontier DP are established ideas. Becker [4] gives a sequential mixed-weight DP for a near-minimal *height* restriction in quadratic time. The band above uses excess *total unweighted path length*, a different restriction; it does not improve Becker's stated problem. Priority of this particular potential, band bound, and exclusion combination remains unestablished.

8 Exact checks and preliminary measurements

The code uses nonnegative integer weights, an independent cubic recurrence, Knuth's recurrence, and direct evaluation of reconstructed trees. The basic harness additionally enumerates all tree shapes for small sizes. CPU experiments ran on the sponsor's M4 Mac mini. The reported quantities are algorithmic counts, not inferred asymptotic bounds from elapsed time.

- Single-key contraction: 22,143 exhaustive weight assignments in $\{0, 1, 2\}$ through $n = 4$, and 6,000 seeded random instances through $n = 20$. Every cost and reconstructed tree matched the cubic oracle.
- Block theorem: 19,486 qualifying interval contractions across 4,500 seeded instances through $n = 16$ matched independent cubic costs.
- Block algorithm: 3,000 random instances through $n = 20$, including fixed and adaptive thresholds, matched the cubic oracle and tree evaluation. Additional small cases check threshold selection against every threshold.

Table 1 records one seeded instance per family at $n = 512$. The baseline has 131,328 states. The block column includes all states inside selected blocks, not just the core. Results from different solvers use their separately recorded seeds and are not paired timing experiments.

Table 1: Preliminary exact DP-state counts, including block solves.

Family	Single-key pass + DP	Adaptive blocks + DP
Uniform mixed weights	131,328	131,328
Near-uniform mixed weights	131,328	131,328
Random mixed weights	128,778	130,817
Log-uniform weights	116,886	94,964
Keys only (control)	92,235	88,508
Sparse heavy separators	—	10,344

In the heavy-separator instance all gaps and ordinary keys have integer weights in $\{1, 2, 3\}$, and 22 randomly selected keys have weight 10^6. The block algorithm retains exactly 22 keys. Its state reduction is about 12.7-fold, but this is not a claim of a 12.7-fold runtime speedup. On ordinary random mixed weights the gain is small. The single-key pass contracts every key for the tested sequence $p_i = 4^i, q_i = 4^i$, but not for most of the ratio-two sequence. Geometric integer inputs also illustrate why bit complexity must be distinguished from arithmetic-operation complexity.

9 A useful relaxation that failed as a general solver

Allow arbitrary assignment of key weights to internal positions and gap weights to external positions. In this relaxed problem, repeatedly take the two smallest current external weights a, b and the smallest remaining internal weight p, replace them by $a + b + p$, and add that sum to the cost.

To prove the rule, take a deepest internal node of an optimum. Its children are leaves. Swapping the smallest internal weight into it and the two smallest external weights into its children cannot increase cost. Contract that cherry and induct. Sorting and a heap give $O(n \log n)$ time. This elementary extension of the Huffman exchange argument is not claimed as novel.

The relaxed cost is a valid lower bound for full OBST. Equality with a feasible ordered tree certifies optimality. It agreed with an independent shape-enumeration and rearrangement oracle on 22,143 exhaustive and 2,700 random instances in the basic harness.

However, using the relaxation on demand in an exact sparse DP was poor on random inputs. At $n = 128$ it visited roughly 4,300–4,600 exact intervals instead of Knuth's 8,256, but spent approximately 3.9–4.1 million weight comparisons constructing bounds and inspected over 100,000 candidate roots. Fewer exact states did not mean less computation.

Nor does the relaxation's interval matrix inherit the required quadrangle inequality. For $p = (9, 1, 12)$, $q = (5, 1, 18, 10)$, its costs H satisfy

$$H(1,2) + H(2,3) = 60 > 54 = H(1,3) + H(2,2).$$

Thus applying Knuth-style matrix optimization directly to this bound would be unjustified.

Two additional exact counterexamples exclude tempting local certificates. With $p = (3, 5, 18)$, $q = (13, 14, 4, 8)$, the tree with root 2 and left subtree $0(\varnothing, 1)$ costs 127, every single-rotation neighbor costs at least 134, yet the balanced tree costs 125. With $p = (18, 8, 18)$, $q = (17, 11, 1, 13)$, the balanced tree costs 164 and every distinct tree obtained by promoting a descendant to the root of an existing subtree costs at least 169; the tree $0(\varnothing, 2(1, \varnothing))$ costs 157. These are barriers to those local certificates, not complexity lower bounds.

10 An ordered depth-chain lower bound

Let l_i be the comparison length of key i and g_i the depth of gap i. For every valid BST, adjacent keys have different lengths and

$$g_0 = l_0, \quad g_n = l_{n-1}, \quad g_i = \max(l_{i-1}, l_i) \quad (0 < i < n).$$

This identity is existing work; see Becker [4], Lemma 1. Let K_d count keys of length d and G_d count gaps at depth d. The root and child-count identities are

$$K_1 = 1, \qquad G_d + K_{d+1} - 2K_d = 0 \quad (1 \le d \le n), \qquad K_{n+1} = 0.$$

Permit any sequence in $\{1, \dots, n\}^n$ with unequal adjacent entries. For arbitrary multipliers $\lambda_0, \dots, \lambda_n$, minimize

$$\sum_i p_i l_i + \sum_i q_i g_i + \lambda_0(K_1 - 1) + \sum_{d=1}^{n} \lambda_d(G_d + K_{d+1} - 2K_d). \tag{3}$$

Call the minimum $L(\lambda)$. Every actual tree is permitted and has zero penalty, so $L(\lambda) \le \mathrm{OPT}$ for every multiplier vector. The expression is a chain with key energy $U_i(b) = p_i b + \lambda_{b-1} - 2\lambda_b$, gap energy $V_i(b) = q_i b + \lambda_b$, and constant $-\lambda_0$. For an interior transition into length b, take the smaller of

$$\min_{a<b} F(a) + V_i(b), \qquad \min_{a>b} \{F(a) + V_i(a)\}.$$

Prefix and suffix minima give $O(H)$ work per key for a domain of H lengths. Using $H = n$ would cost quadratic time even for one bound evaluation.

Theorem 6 (Exact affine-tail compression)**.** *Fix $h \ge 1$, arbitrary $\lambda_0, \dots, \lambda_h$, and a slope $0 \le s \le \min_i p_i$. Extend $\lambda_d = \lambda_h + s(d - h)$ for $d > h$. The minimum in (3) is unchanged if lengths are restricted to $1, \dots, \min(n, h + 3)$.*

Proof. For $b \ge h + 1$, the unary key energy has slope $p_i - s \ge 0$ and the gap energy has slope $q_i + s \ge 0$. Both are nondecreasing in this tail. If some $l_i > h + 3$, replace it by a member of $\{h + 1, h + 2, h + 3\}$ unequal to its current neighbors. At most two values are forbidden. The key energy does not increase. Each adjacent maximum is either unchanged or decreases within the affine tail, so neither gap energy increases. Endpoint gaps obey the same argument. Repeat for all large lengths. The restricted domain is a subset of the original domain, and every original sequence has a no-more-expensive restricted sequence; therefore the minima are equal. □

Taking $h = \lceil \log_2(n+1) \rceil + 2$ gives an $O(n \log n)$ oracle. This is a statement about minimization of the relaxed energy, with no height assumption on the unknown optimal BST. Optimizing only the head multipliers requires folding all tail constraint violations into the derivative for λ_h. The implementation uses exact integer multipliers and costs scaled by 1024. Its subgradient ascent is heuristic; the bound is valid at every iteration regardless of convergence.

A stronger oracle requires exactly one length1 key. For each candidate root, the left and right chains then contain only lengths at least2. One forward and one reverse pass supply their closing costs for every root position, giving all forced-root bounds in $O(nH)$ total time. The projection above preserves the unique root, so tail compression remains valid for each marginal. Retaining the maximum bound seen for each root, then taking the minimum over roots, is also a valid lower bound.

Both oracles passed 50 comparisons against exhaustive length-sequence minimization, 5,000 arbitrary-multiplier bound checks against cubic OBST, and 1,050 direct comparisons of the full and shortened oracles. Forced-root marginals were additionally checked against exact root-constrained costs.

A structural gap. The sequence $l = (3, 1, 2, 3, 2)$ has $g = (3, 3, 2, 3, 3, 2)$, a unique root, and all the level-count equalities. It is not a valid BST depth sequence. For

$$p = (3, 27, 21, 16, 12), \qquad q = (18, 22, 27, 22, 12, 22),$$

it costs 470, while the true optimum is 472. Its penalty is zero for every multiplier vector. Hence neither oracle family can ever exceed 470 on this instance. More optimization iterations cannot remove this obstruction.

Preliminary strength and cost. At $n = 256$, normalized gaps $(\mathrm{OPT} - \mathrm{LB})/W$ for the chain bound versus order-free Huffman were 0.15003 versus 0.22080 on a random mixed instance, and 0.00873 versus 0.49294 on an alternating instance. However, 300 ascent iterations used 1,082,368 key-depth cells including initialization, compared with 32,896 interval states for Knuth. The unique-root variant uses two passes and still leaves 254 of 256 potentially improving roots on the random instance, and 255 on the alternating instance. Only two survive on the gap-spike instance. These results identify a lower-bound component and its limitations; they do not establish a faster general exact solver.

11 Exact depth ordering and an arbitrary-window obstruction

Call a positive integer sequence *ranked* if every nonempty contiguous interval has a unique minimum. Equivalently, equal labels have a smaller label between them. This is vertex ranking on a path with the label order reversed; vertex ranking and bounded-path variants are existing notions [5, 6]. The following results explain the ordering information absent from the preceding relaxation; no novelty claim is made for this characterization.

Theorem 7 (Depth characterization)**.** *A sequence $l \in \{1, \ldots, n\}^n$, $n \geq 1$, is the inorder key-comparison-length sequence of a BST if and only if it is ranked, $K_1 = 1$, and every level-flow equality holds, with gaps derived from l as above.*

Proof. Necessity follows from the unique shallowest key in each inorder interval and the two children of every internal node. For sufficiency, delete all labels at most d. Let U_d count the remaining

nonempty runs, and let A_d count the deleted labels. If the deleted labels form R runs and include e endpoints, then the number of gaps of length at most d is $A_d - R + e$, whereas $U_d = R + 1 - e$. Thus

$$G_{\le d} = A_d + 1 - U_d.$$

Summing the level-flow equalities from 1 through d, and using $K_1 = 1$, also gives

$$G_{\le d} = A_d + 1 - K_{d+1}.$$

Hence $U_d = K_{d+1}$. Ranking permits at most one label $d+1$ per remaining run, so equality forces exactly one. Starting with the unique length1 root, every remaining nonempty interval therefore receives its root at the next level. This constructs the tree with exactly the supplied lengths. □

Ranking has a linear recognizer: scan with an increasing stack, pop larger labels, reject an equal top, and otherwise push. A rejected pair has no smaller intervening label; conversely every such pair causes rejection. Each entry is pushed once and popped at most once. This establishes cheap recognition, not cheap minimization over all valid sequences.

Theorem 8 (Bounded-window obstruction)**.** *For every integer $w \ge 2$ and $t \ge 2$, there is a strictly positive full-OBST instance with $n = w + 3$ keys and an inadmissible depth sequence satisfying all level-flow equalities, a unique length1 root, and ranking on every interval of at most w keys. Its cost is at most $(7/8 + 1/(8t))\,\mathrm{OPT}$. Consequently every lower bound obtained by pricing the level-flow equalities over this window-constrained domain has the same upper limitation, regardless of the multipliers.*

Proof. Take

$$l = (3, 1, 2, 3, 4, \ldots, w+1, 2).$$

The ascending middle portion has $w - 1$ entries. Its two enclosing length2 labels form the only interval with a repeated minimum, of width $w+1$. Every shorter window passes. Direct substitution into the gap formula verifies $K_1 = 1$ and every level-flow equality. The sequence is invalid: the unique root is key1, but its singleton left subtree has length3, and two keys to its right both claim length2.

Give every gap weight1 and every key weight1 except

$$p_1 = 3B, \qquad p_2 = p_{n-1} = B.$$

The unit-weight outcomes contribute $E = w^2 + 4w + 10$ in the inadmissible sequence, whose cost is $7B + E$. Every actual tree costs at least $8B$. If key1 is the root, the two B-weight keys lie to its right and their comparison lengths sum to at least5, giving $3B + 5B$. Otherwise key1 has length at least2 and each B-weight key has length at least1, giving $6B + 2B$.

Set $B = tE$. The inadmissible cost satisfies

$$7B + E = \left(\frac{7}{8} + \frac{1}{8t}\right) 8B \le \left(\frac{7}{8} + \frac{1}{8t}\right) \mathrm{OPT}\,.$$

Its level-flow penalty is zero for every multiplier vector, so every priced minimum is at most this cost. □

For $t = 16$ the upper ratio is $113/128$, independently of w; as t increases it approaches $7/8$. This rules out universal exactness of the specified window-minimum repair, not other local constraints, adaptive nonlocal cuts, or subquadratic full-OBST algorithms. Equivalently, at every $n \ge 5$ the construction passes all windows of up to $n - 3$ keys.

Burying the obstruction below a forced prefix. Let a current instance have mass W and a feasible tree of cost U. Prepend a key of weight $A = W + U + 2$ and a new leftmost gap of weight1. Using the new key as root costs $A + 1 + W + U = 2A - 1$, while every other root costs at least $2A$ from that key alone. Hence the new root is forced and $\mathrm{OPT}_{\mathrm{new}} = A + 1 + W + \mathrm{OPT}_{\mathrm{old}}$. The transformation $l \mapsto (1, l_0 + 1, \ldots, l_{n-1} + 1)$ preserves zero flow and local window ranking, and changes the inadmissible cost by the same $A + 1 + W$. Repeating this D times places the strict gap beneath the true optimal top-D right-chain prefix. Thus fixing any constant number of top levels cannot make the remaining window/flow relaxation exact. The constant relative gap is not claimed after this embedding, because the forcing weights add a potentially large common cost.

Independent verification. The harness enumerated all 873,612 sequences in $\{1, \ldots, n\}^n$ for $1 \leq n \leq 7$. The characterization accepted exactly the independently generated BST depth sequences: respectively 1, 2, 5, 14, 42, 132, and 429. The stack recognizer agreed with direct interval scans on every sequence. Width3 and exact flow already admit four invalid sequences at $n = 6$ and 34 at $n = 7$. The positive family was checked at $t = 16$ for all 127 widths $2 \leq w \leq 128$: exact costs, window validity, zero flow, and the tree-cost lower bound all passed. Knuth and direct tree evaluation checked every instance; independent cubic DP also checked $w \leq 24$. Another 120 cases combine $2 \leq w \leq 16$ with forced prefix depths $1 \leq D \leq 8$. Cubic DP, Knuth, direct costs, and inspection of the optimal prefix verify the exact additive identity and preserved strict gap.

12 Status and reproducibility

The general subquadratic question remains open. This checkpoint provides exact contraction and bounded-excess constructions, global certificates, and scoped obstructions to depth relaxations. Mathematical novelty, broad practical benefit, and comparisons against Larmore-style implementations remain to be assessed.

Standard-library Python code, seeded exact results, and matched timing observations are under `obst/code` and `obst/results`. The README gives reproduction commands. The active objective remains to avoid most of the quadratic table on a substantially broader class, or establish a rigorously scoped complexity lower bound.

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